Deep Learning

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Outline

- Fundamentals of Deep Learning

Instructor

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- Research Interest: Graph Data Science, Social Network Analysis, Graph Mining

Deep Learning

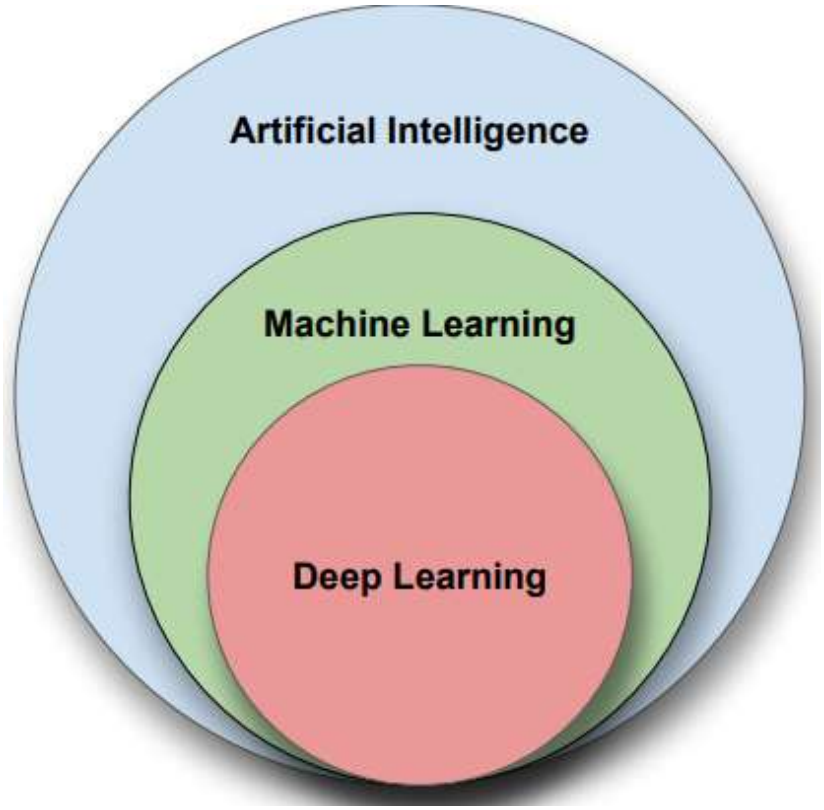
- **AI** is intelligence demonstrated by machines, as opposed to natural intelligence displayed by animals including humans.

- **ML** is the study of computer algorithms that can improve automatically through experience and using data. It is seen as a part of artificial intelligence.

- **DL** is part of a broader family of machine learning methods based on artificial neural networks.

Deep Learning

- AI is anything about man-made intelligence exhibited by machines.
- ML is an approach to achieve AI.
- DL is one technique to implement ML.



Deep Learning

- Deep Learning is a subset of Machine Learning that uses mathematical functions to map the input to the output.

- These functions can extract non-redundant information or patterns from the data, which enables them to form a relationship between the input and the output.

- This is known as learning, and the process of learning is called training.

Deep Learning

- Modern deep learning models use artificial neural networks or simply neural networks to extract information.
- These neural networks are made up of a simple mathematical function that can be stacked on top of each other and arranged in the form of layers, giving them a sense of depth, hence the term Deep Learning.

- Deep Learning was first theorized in the 1980s, but it has only become useful recently because:
  - It requires large amounts of labeled data.
  - It requires significant computational power (high performing GPUs)

Neural Networks

- A Neural Network (NN) is a network of neurons that are used to process information.
- A neural network is a group of interconnected units called neurons that send signals to one another. Neurons can be either **biological cells** or **mathematical models**.

- The most advanced data processing machine at the time — the brain.

Types of Neural Networks

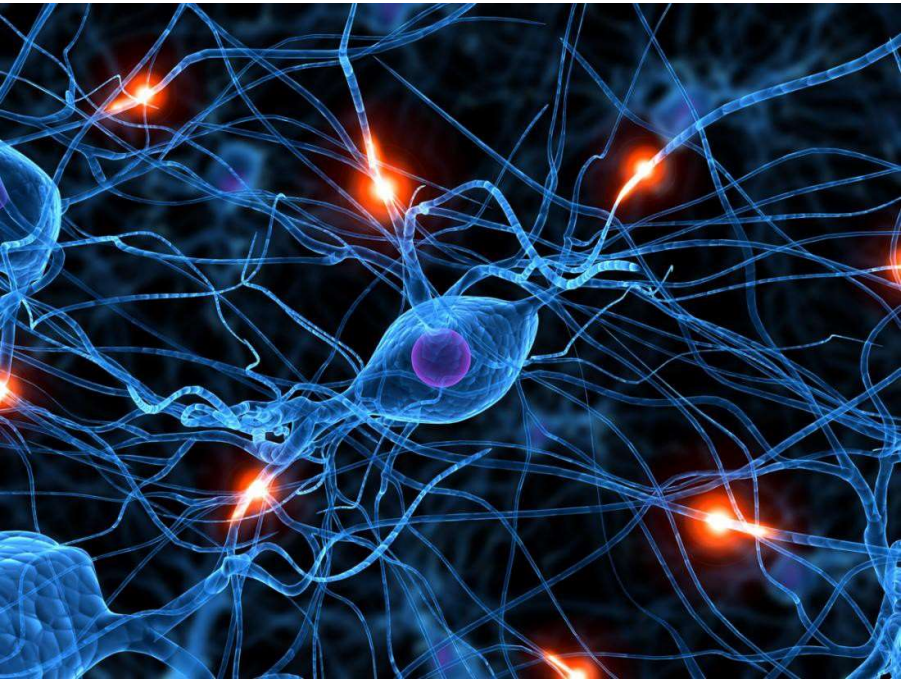
- In neuroscience, a **Biological Neural Network (BNN)** is a physical structure found in brains and complex nervous systems – a population of nerve cells connected by synapses.

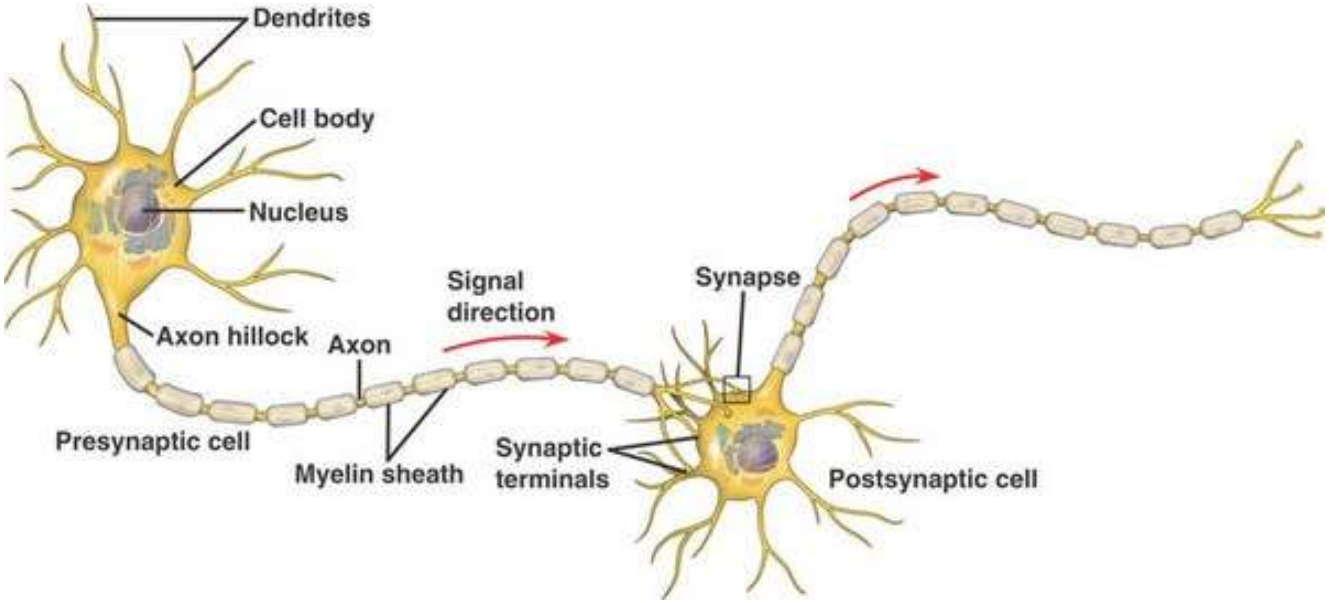
- In machine learning, an **Artificial Neural Network (ANN)** is a **mathematical model** used to approximate nonlinear functions. Artificial neural networks are used to solve artificial intelligence problems.

Biological Neural Networks

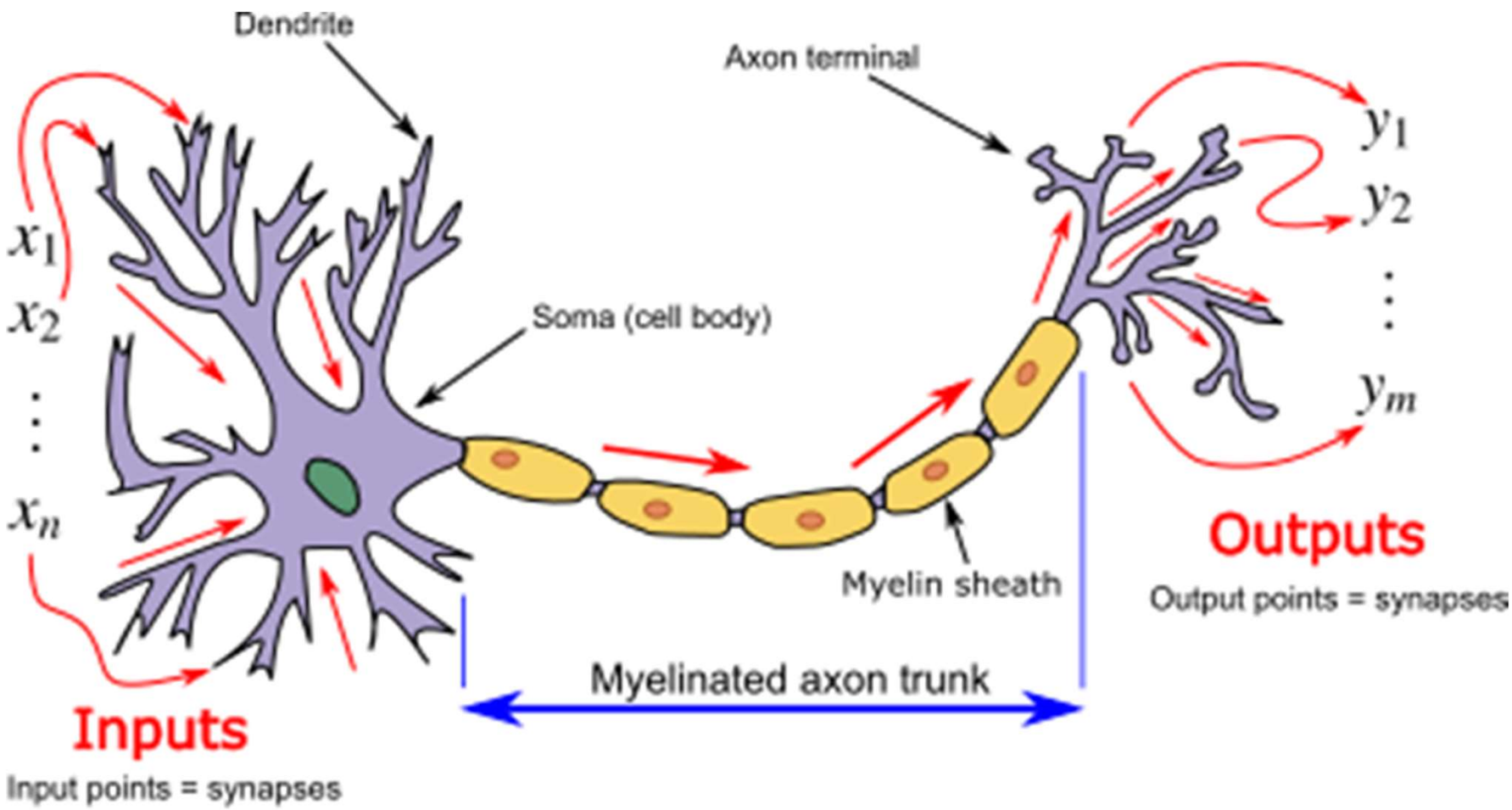
- In the context of biology, a neural network is a population of biological neurons chemically connected to each other by synapses.
- A given neuron can be connected to hundreds of thousands of synapses.
- Each neuron sends and receives electrochemical signals called action potentials to its connected neighbors.
- A neuron can serve an excitatory role, amplifying and propagating signals it receives, or an inhibitory role, suppressing signals instead.

Biological Neural Networks

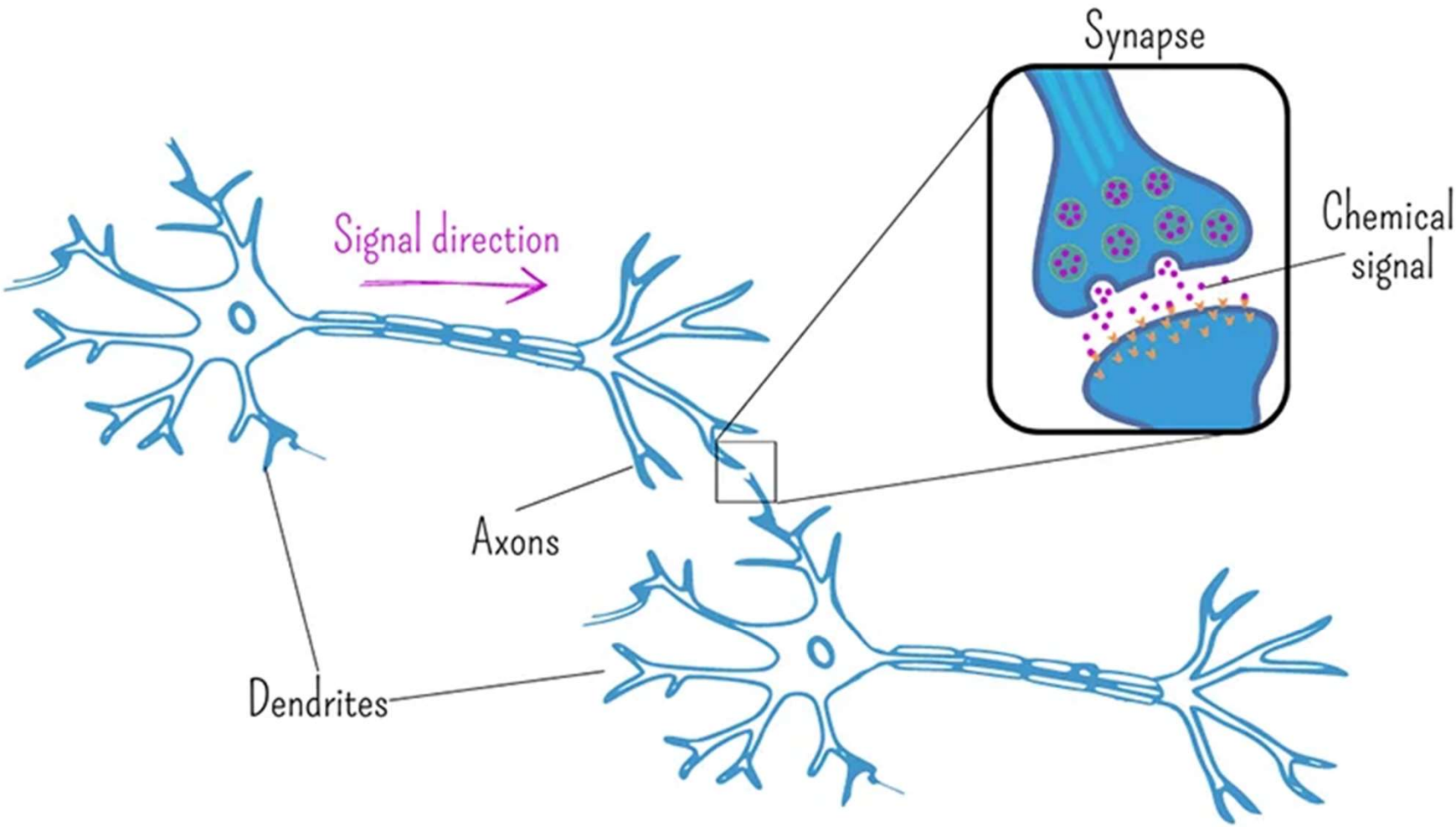




Biological Neural Networks



Biological Neural Networks



Biological Neural Networks

- The brain is principally composed of about 10 billion neurons, each connected to about 10,000 other neurons.
- Each neuron receives electrochemical inputs from other neurons at the dendrites.
- If the sum of these electrical inputs is sufficiently powerful to activate the neuron, it transmits an electrochemical signal along the axon and passes this signal to the other neurons whose dendrites are attached at any of the axon terminals. These attached neurons may then fire.
- It is important to note that a neuron fires only if the total signal received at the cell body exceeds a certain level. The neuron either fires or it doesn't, there aren't different grades of firing.

Types of Neural Networks

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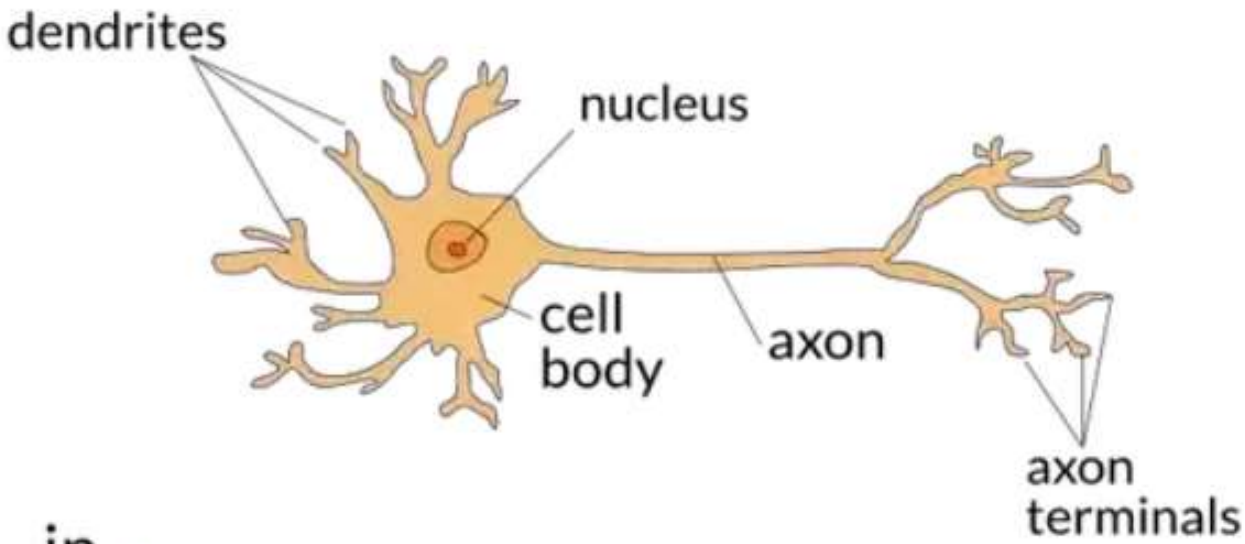
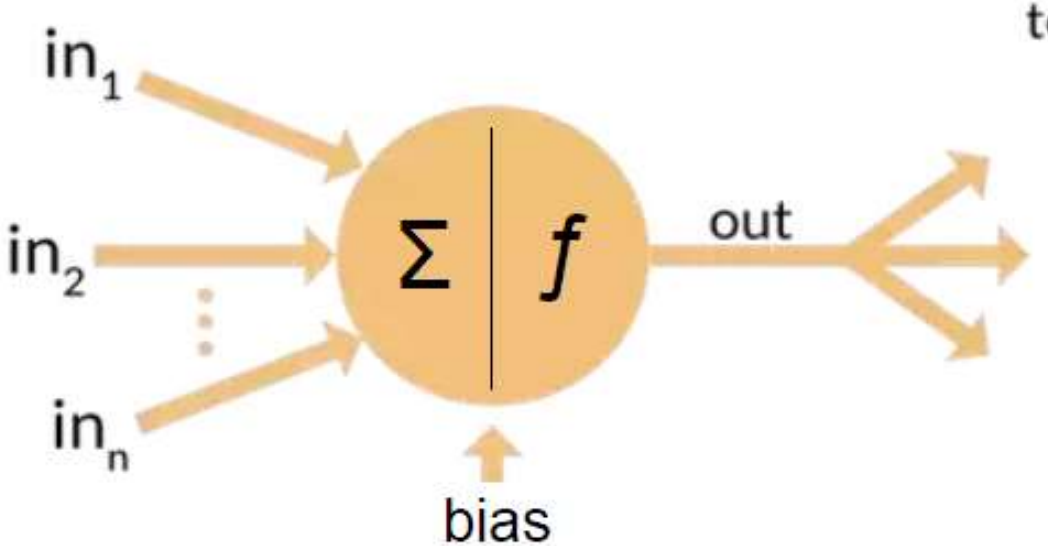
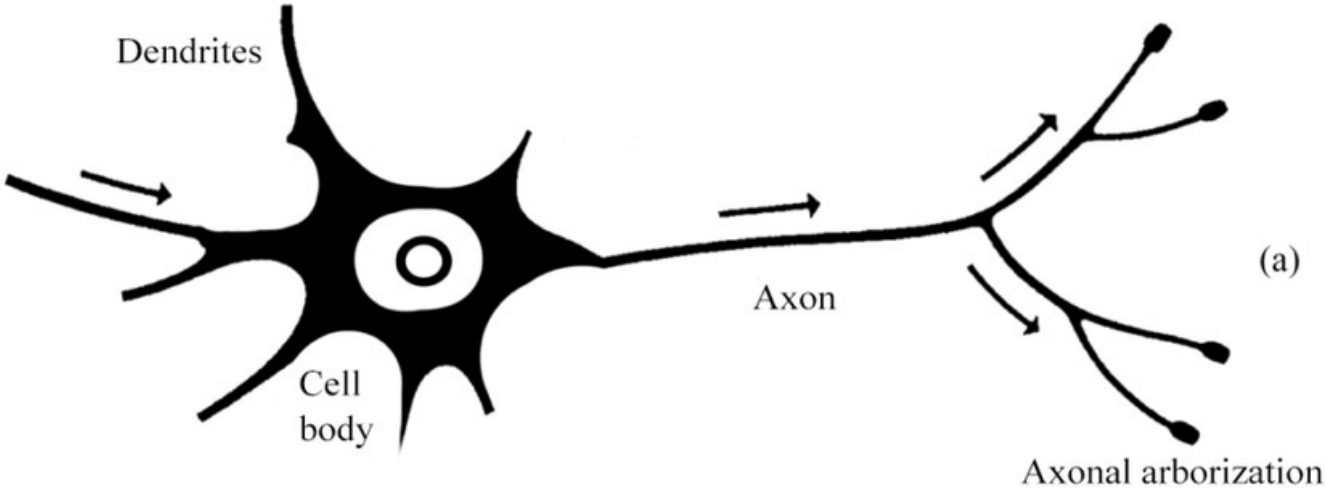
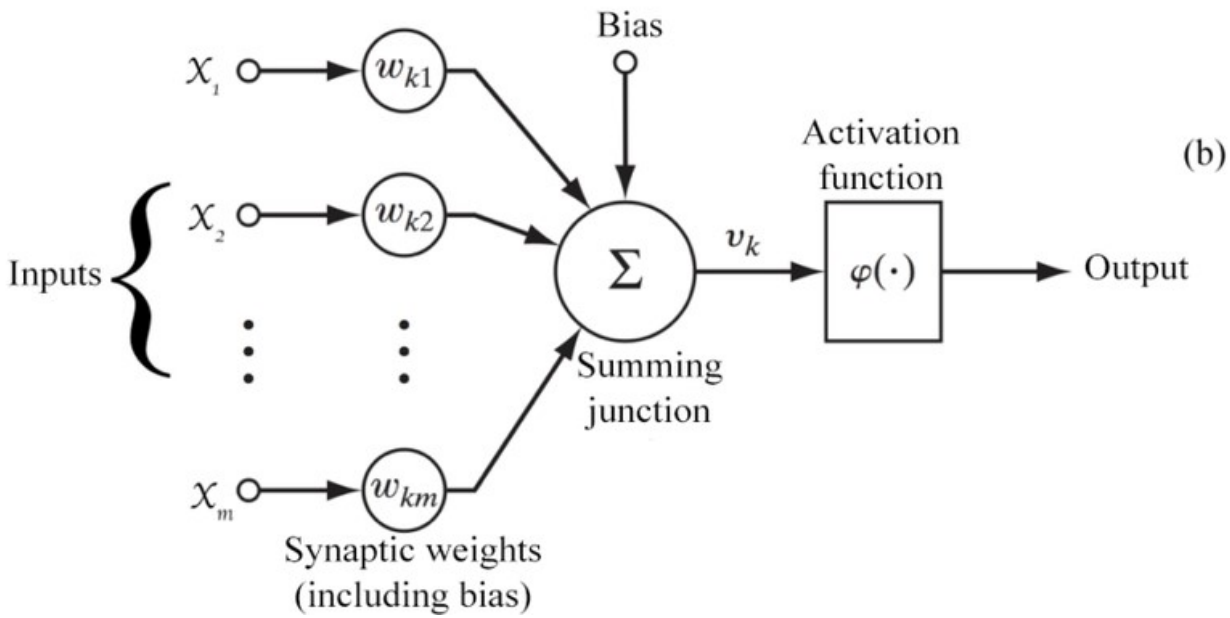
Neural Networks





Artificial Neural Networks





Artificial Neural Networks

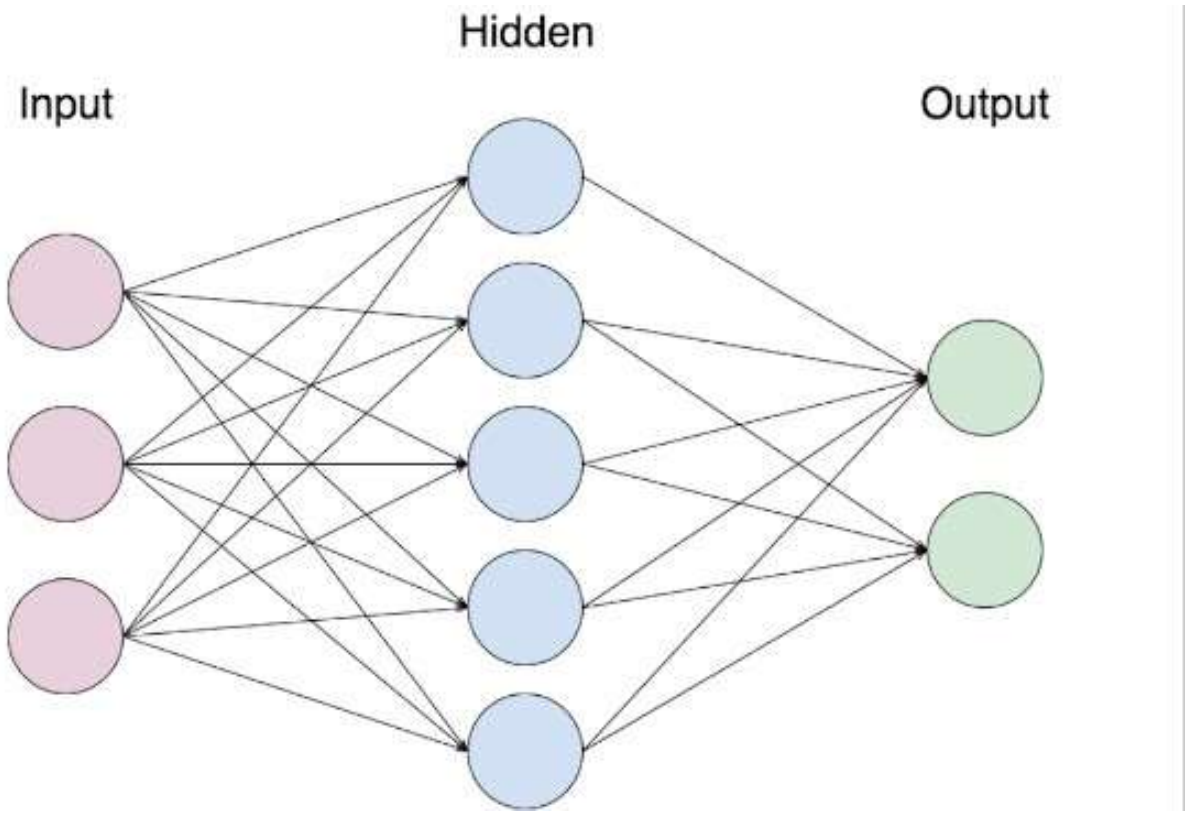
- An artificial neural network (ANN) is composed of artificial neurons or nodes.
- The connections of the biological neuron are modeled in artificial neural networks as weights between nodes.
- A positive weight reflects an excitatory connection, while negative values mean inhibitory connections.

Artificial Neural Networks

- An ANN consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain.
- These are connected by edges, which model the synapses in the brain.
- Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons.
- The "signal" is a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs, called the activation function.
- The strength of the signal at each connection is determined by a weight, which adjusts during the learning process.

Architecture of Simple Neural Networks

- An Artificial Neural Network is made up of 3 components:



Architecture of Simple Neural Networks

- **Input Layer:** This is literally the layer that inputs information for the neural network to process. Each circle represents 1 feature (a piece of information).

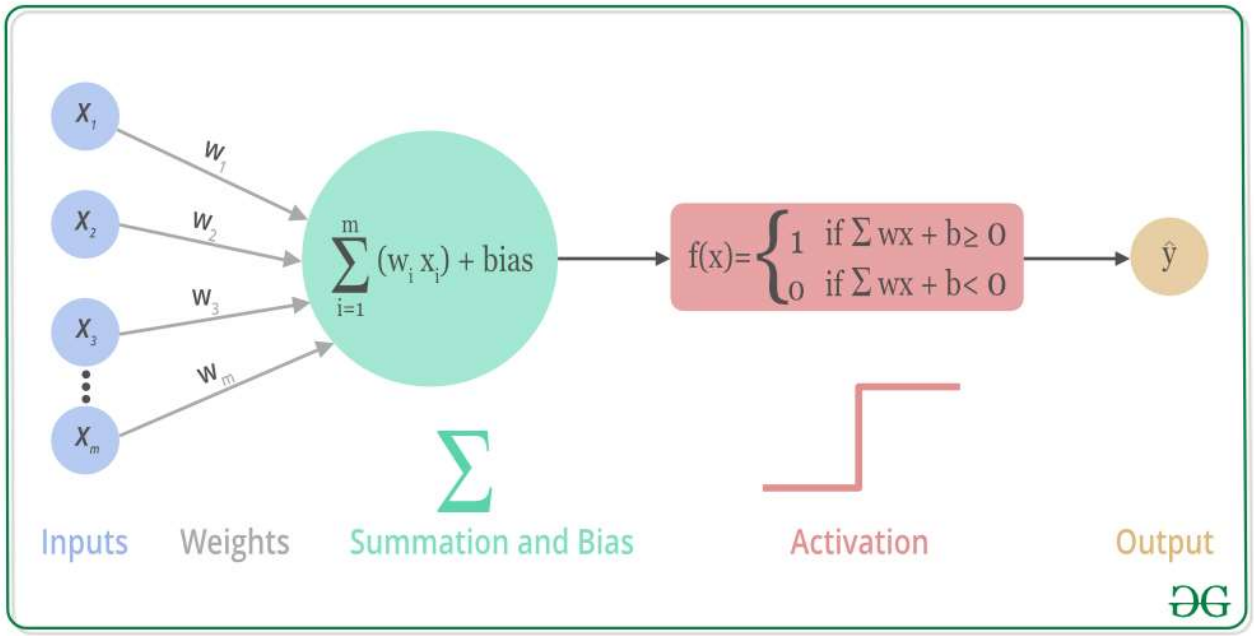
- **Hidden Layer:** These layers do all the processing for neural networks. Each layer consists of nodes that mimic our brains' neurons. These nodes receive information from the previous layer's nodes, multiply it by **weight** and then add a **bias** to it.

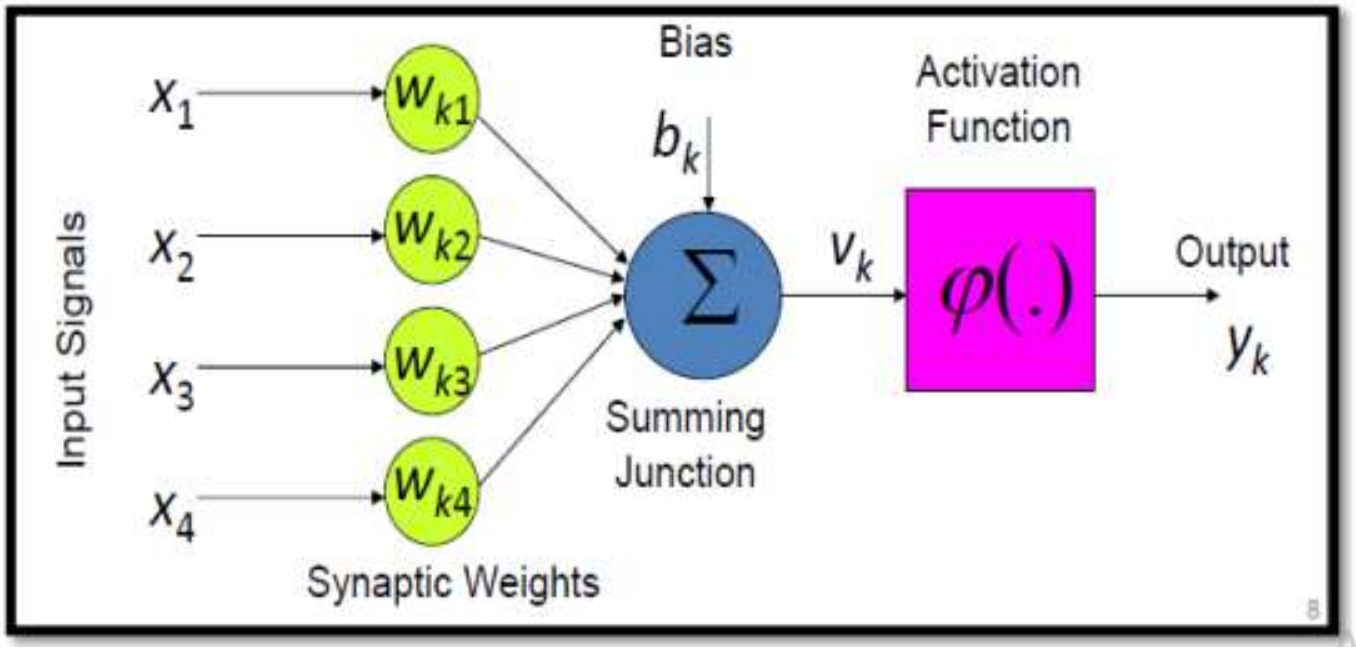
- **Output Layer:** The output layer is the output/conclusions of the model derived from all the computations performed. There can be single or multiple nodes in the output layer.

Artificial Neural Networks

- A neural network is characterized by
  - Its pattern of connections between the neurons (called its architecture).
  - Its method of determining the weights on the connections (called its training, or learning, algorithm).
  - Its activation function.

Artificial Neural Networks





Summary

- Fundamentals of Deep Learning